Title:

Sweet Sales Dashboard-
Understanding Patterns in Candy Distribution

Dashboards with Tableau - ISBA 2404

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Overview of the Problem

Candy distributors face challenges in managing high volumes of orders, diverse product offerings, and varying customer segments. Our project aims to analyze historical sales, shipping, and product performance data to provide actionable insights that enhance both profitability and customer satisfaction.

The central question we address is: **"How can a candy distributor optimize sales and delivery performance across different customer segments, regions, and product categories?"**

This issue is particularly relevant for operations managers, logistics teams, and regional sales leaders, who must balance demand with supply while minimizing costs and maximizing revenue. Our dashboard visualizes key metrics to support strategic decision-making around shipping efficiency, regional profitability, and product-level trends.

Analysis of the Data

The data is acquired from the Maven Analytics website, detailing various attributes from the candy distributor's sales records.

The dataset has various columns like:

- Order ID – unique order identifier
- Order Date – date the order was placed
- Ship Date – date the order was shipped
- Ship Mode – shipping method used
- Customer ID and Customer Name – customer-level identifiers
- Segment – customer category (e.g., Corporate, Consumer)
- Region, State, and City – geographic sales data
- Product ID, Category, Sub-Category, Product Name – item-level sales data
- Sales, Quantity, Discount, Profit – financial and volume metrics

Tables Used and Relationships:
There are three main tables:

1. **Candy_Factories.csv** – Contains factory names and geographic coordinates (latitude, longitude).
2. **Candy_Products.csv** – Lists product details, pricing, and the associated manufacturing factory.
3. **Candy_Sales.csv** – Historical order data including order/shipping dates, regions, products, customer info, and key sales metrics.

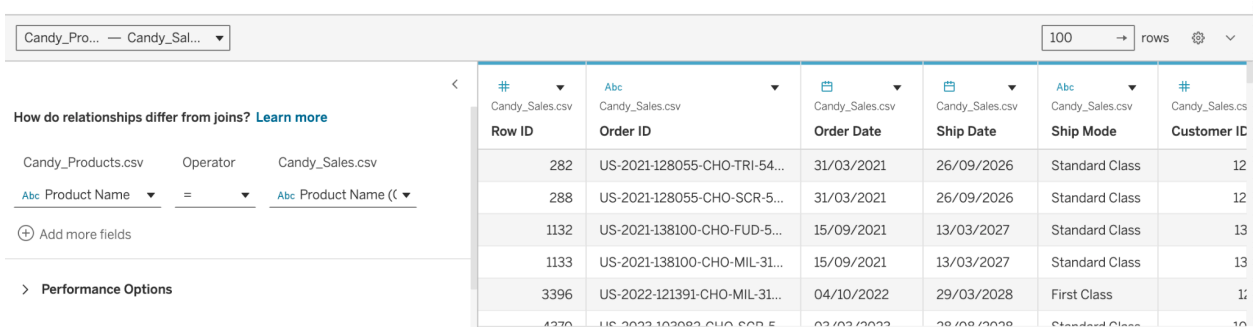
Data Preprocessing and Joins:

- The datasets were cleaned in Tableau. No major nulls were present.
- **Data Interpreter** in Tableau was used for initial data profiling.

We implemented the following joins:

- Candy_Factories → Candy_Products using the Factory Name field
- Candy_Products → Candy_Sales using the Product Name

These joins allowed us to build a unified data model connecting sales performance to factory output and product attributes.



Calculated Fields And Parameters:

- Standardized product and region names.
- Created calculated fields for Avg. Order Value, Stability Metrics.
- Created parameter for selected candies.

Key Focus Areas for Analysis

Our analysis will focus on identifying patterns and insights that help the distributor improve overall sales and operational efficiency. Key areas we're currently exploring include:

- Shipping performance and delivery times across regions
- Regional and segment-based trends in sales and profitability
- Seasonal variations and demand peaks
- Product performance based on quantity sold, revenue, and discount impact

Key Dashboards and Insights

Dashboard 1

- **KPI Summary Cards**
  ➢ Displays key metrics: Total Production, Customer Count, Total Sales, and Avg Order Value.

- **Sales Performance by State (Map)**
  ➢ Geographic map showing where sales are highest across U.S. states.

- **Top Products by Sales (Bar Chart)**
  ➢ Visualizes the top-selling candy products by total units sold.

- **Regional Sales Distribution (Pie Chart)**
  ➢ Shows percentage share of total sales across regions (Atlantic, Gulf, Pacific, Interior).

- **Heatmap - Region vs Product Type**
  ➢ Cross-analysis of which products are popular in which regions using color intensity and sales count.

Dashboard 2

- **KPI Summary Cards**
  ➤ Repeats key metrics for context: Total Production, Customer Count, Total Sales, and Avg Order Value (2021–2024).

- **Sales and Profit by Month**
  ➤ Combo chart showing monthly sales (bar) and gross profit % (line), revealing seasonal patterns and margin shifts.

- **Top-Selling Products by Sales Contribution (Pareto Chart)**
  ➤ Highlights how a few products drive the majority of sales, based on the 80/20 principle.

- **Candy Type Stability Analysis (Scatter Plot)**
  ➤ Compares each product's average monthly sales vs. sales variability to identify reliable vs. risky performers.

Overall Business Insights

1. **Geographic Performance Disparity**
   Certain states like California and New York significantly outperform others in both revenue and profit. This indicates potential market saturation in high-performing regions and growth opportunities in underperforming ones.
2. **Seasonal Demand Trends**
   Sales consistently peak in Q4, likely due to holiday demand. This trend is evident across product categories and regions, signaling an opportunity for seasonal promotions and inventory adjustments.
3. **Customer Distribution vs. Revenue**
   Some regions have a high number of customers but low total revenue, suggesting either smaller order sizes or preference for lower-margin products.
4. **Mismatch Between Unit Sales and Profit**
   Products with high unit sales but low profit margins indicate areas for price review or cost optimization (e.g., supply chain, packaging).

5. **Top-Selling Products Dominate Portfolio**
   A small number of candy products, especially in the Chocolate division, account for a large portion of total sales, demonstrating the classic 80/20 Pareto trend.

Strategic Recommendations

1. **Focus on High-Profit Regions**
   Prioritize marketing and logistics investments in regions that generate high profit, while reassessing efforts in low-ROI areas.
2. **Adjust Pricing Strategies**
   For products with high unit sales but low profitability, test price increases or cost-reduction initiatives to improve margins.
3. **Optimize Seasonal Campaigns**
   Leverage Q4 trends by planning seasonal bundles, discounts, and inventory ramp-ups ahead of holiday demand.
4. **Expand Popular Products into Untapped Regions**
   Target mid-performing regions with the top 3 best-selling products, supported by localized campaigns to boost adoption.
5. **Introduce Customer Segmentation-Based Promotions**
   Use regional and customer insights to offer tailored discounts or loyalty programs to boost repeat purchases and retention.
6. **Monitor Low-Converting Regions**
   Identify and explore barriers in underperforming regions, such as delayed shipping, weak local distribution, or lack of product variety.

Narrative Elaboration

Visualization Approach

We used a combination of KPI tiles, bar charts, line graphs, and regional maps to address our core business questions. Each chart was selected based on the metric being analyzed — e.g., line charts for trends, bar charts for comparisons, and maps for spatial distribution. This design choice ensures that decision-makers can quickly grasp key metrics related to geography, time, and product performance.

Calculated Fields

We created custom calculated fields such as:

- Avg order value= SUM(Sales) / COUNTD(OrderId)
- Gross Profit = Sales - Cost
- Profit Margin = (Gross Profit / Sales) These were used to analyze efficiency and profitability across regions and products.
- Avg Monthly Sales = { FIXED [Product Name], MONTH([Order Date]) : AVG([Sales]) }
- Sales Variability (Standard Deviation) = { FIXED [Product Name], MONTH([Order Date]) : STDEV([Sales]) }

Parameters

A key parameter for selected candy (Is Selected Candy?)was implemented to make the dashboard dynamic and customizable based on user needs.

Dashboard Interactivity

We employed Dashboard Actions such as:

- Filter actions to drill down from state-level to product-level data
- Highlight actions to emphasize trends over time or specific regions
- Parameter controls to update views based on user input